RATIONAL EXPONENTS

Combining radicals with different indices

Direct combination is impossible when the indices differ. Converting to exponents provides a path.

01

DIFFERENT INDICES — BEFORE

$$\sqrt{2}\ \sqrt[3]{4}$$

Convert — to exponents over one base: $4^{1/3} = (2^2)^{1/3} = 2^{2/3}$

02

EXPONENT FORM, ONE BASE

$$2^{1/2} \cdot 2^{2/3}$$

Add exponents — 3/6 + 4/6 = 7/6

03

SINGLE POWER

$$2^{7/6}$$

Convert back — the denominator 6 becomes the index

04

SINGLE RADICAL

AFTER

$\sqrt[6]{128}$

DIVISION